

Numerical and Approximate Solution Methods for the Signature (4,4) Einstein-Spinor Components on $\{\mathbf{x}_0, \dots, \mathbf{x}_7\}$

Complete solver-method provenance and dark-sector interpretation report

Reproducible exact-real implementation

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Abstract

This report verifies the numerical and approximate methods for the 18-equation homogeneous reduction of the Einstein-spinor model written in

`coordinates = {x0, x1, x2, x3, x4, x5, x6, x7}`

with evolution variable `x4`. The external bridge notebook supplies only these coordinate names and their ordering; its metric and connection are not used. The selected production method is the pinned safe-Rust SUNDIALS 7.8.0 CVODE BDF implementation. A separate exact reduction to one background quadrature and eight spinor rotations supplies analytic structure and asymptotic approximations. No comparison study establishes BDF as globally optimal.

1 Exact initial-value problem solved

State vector:

$$y(x4) = (a, H, \psi_1, \dots, \psi_{16}) \in \mathbb{R}^{18}.$$

The report uses one-based spinor labels. They map to zero-based code and CSV columns by `psi_A = psi(A-1)`. Define

$$U = \frac{1}{20} + \frac{19}{100}S^{-4/5}.$$

The geometry equations are

$$\dot{a} = aH, \quad \dot{H} = -\frac{21}{6}(\rho + p).$$

For each $j=1, \dots, 8$, the two spinor equations are exactly

$$\dot{\psi}_j = -\frac{7}{2}H\psi_j - U\psi_{j+8}, \quad \dot{\psi}_{j+8} = -\frac{7}{2}H\psi_{j+8} + U\psi_j.$$

This represents all 16 scalar equations; they are listed individually in the companion component report. The closures are

$$S = \psi^\top C \psi, \quad \rho = \frac{1}{20}S + \frac{19}{20}S^{1/5},$$

$$p = -\frac{19}{25}S^{1/5}, \quad H^2 = \rho.$$

The reference state is imposed at `x4=0`:

$$a = 1, \quad H = 1, \quad \psi_1 = 1, \quad \psi_{16} = \frac{1}{2}, \quad \psi_A = 0 \text{ otherwise.}$$

CVODE integrates independently from this state backward to -0.2 and forward to 1.5 ; the backward branch is reversed before both branches are joined.

2 Selected production method and exact solver configuration

The implemented production method is variable-step, variable-order BDF in the safe-Rust port of SUNDIALS CVODE 7.8.0. The solver submodule commit is frozen in the machine-readable evidence file.

Source inspection and executable evidence establish this configuration:

1. `CVodeCreate(CV_BDF)` selects BDF.
2. CVODE starts at order one and permits orders one through five.
3. `CVodeInit` installs the default Newton nonlinear solver, with at most three nonlinear corrector iterations per attempt in the pinned implementation.
4. The application attaches an 18-by-18 dense matrix and dense direct linear solver.
5. No user Jacobian is supplied, so the internal dense difference-quotient Jacobian is used.
6. A direct dense solve needs no iterative preconditioner; none is configured.

Canonical solver settings:

- Interval: `x4` in $[-0.2, 1.5]$.
- Output step: `0.01`.
- Relative tolerance: `1e-11`.
- Absolute tolerance: `1e-13`.
- Maximum internal step: `0.002`.
- Maximum allowed steps per branch: `1,000,000`.

Canonical run summary:

- Samples: `171`.
- Internal solver steps: `1372`.
- RHS evaluations: `1486`.
- Acceleration transition at `x4=-0.138005293303244986`.

BDF is a defensible robust choice for the nonlinear, potentially rapidly rotating spinor system, but this repository has not benchmarked Rosenbrock, DIRK, Adams, or symplectic alternatives. Therefore “selected method” is the verified claim; “globally best method” is not.

3 Exact reduction and approximate analytic solution

Skew-adjointness of γ_4 with respect to $\mathbb{C}$ gives

$$\dot{S} = -7HS, \quad S(0) = 1, \quad \boxed{S = a^{-7}}.$$

The expanding background therefore reduces exactly to

$$\rho(a) = \frac{1}{20}a^{-7} + \frac{19}{20}a^{-7/5}, \quad H(a) = +\sqrt{\rho(a)},$$

and one scalar quadrature:

$$x_4 - x_{4_0} = \int_{a_0}^a \frac{d\tilde{a}}{\tilde{a}\sqrt{(1/20)\tilde{a}^{-7} + (19/20)\tilde{a}^{-7/5}}}.$$

For the spinor, define $\chi = \hat{a}^{(7/2)} \psi$ and

$$\eta(x_4) = \int_0^{x_4} U(S(s)) ds.$$

Then

$$\dot{\chi} = -U\gamma^4\chi, \quad \chi(x_4) = \left(\cos\theta I_{16} - \sin\theta \gamma^4\right) \chi(0).$$

Thus every pair $j, j+8$ has the explicit reconstruction

$$\begin{aligned} \psi_j &= a^{-7/2} (\cos\theta \psi_j(0) - \sin\theta \psi_{j+8}(0)), \\ \psi_{j+8} &= a^{-7/2} (\sin\theta \psi_j(0) + \cos\theta \psi_{j+8}(0)). \end{aligned}$$

This reduces the approximate solution problem to quadrature for $\mathbf{a}$ and $\boldsymbol{\theta}$. It also supplies an independent structural check on all 16 numerical components.

Two useful single-term asymptotics are:

1. Linear-term dominance, with $m=1/20$:

$$a(x_4)^{7/2} \simeq a_r^{7/2} + \frac{7}{2}\sqrt{m}(x_4 - x_{4_r}).$$

1. Fractional-term dominance, with $\lambda=19/20$:

$$a(x_4)^{7/10} \simeq a_r^{7/10} + \frac{7}{10}\sqrt{\lambda}(x_4 - x_{4_r}).$$

The late expression corresponds to a proportional to $x_4^{(10/7)}$ after an appropriate origin shift and is accelerating. These are regime-specific approximations; the canonical output integrates the full two-term system.

4 Numerical approximation and error control

The approximation strategy is fully controlled, not ad hoc:

1. Integrate the full coupled ODE once with canonical tolerances.
2. Repeat integration with identical settings to enforce byte-level replay.
3. Integrate a refined run with tighter controls (`rtol=1e-12`, `atol=1e-14`, `max_step=0.001`).
4. Compare canonical and refined states componentwise.
5. Recompute every derived quantity from raw state output independently.

Independent consistency checks applied to each equation family:

1. Friedmann closure check from the 44 Einstein component.
2. Pressure/acceleration consistency from the spatial Einstein components.
3. Five-point finite-difference residual checks on all 18 ODE components.
4. Condensate power-law consistency:

$$S(x_4) \sim a(x_4)^{-7}.$$

1. Deterministic replay byte-identity for history and summary artifacts.

For nonzero expected value `q_ref`, define

$$\text{rel}(q, q_{\text{ref}}) = \frac{|q - q_{\text{ref}}|}{\max(|q_{\text{ref}}|, 10^{-300})}.$$

The three invariant metrics are explicitly

$$\varepsilon_S = \max \text{rel}(S, a^{-7}),$$

$$\varepsilon_\rho = \max \text{rel} \left(\rho, \frac{1}{20} a^{-7} + \frac{19}{20} a^{-7/5} \right),$$

$$\varepsilon_F = \max \text{rel}(H^2, \rho).$$

The canonical run records

$$\varepsilon_S \leq 7.44192091691309166 \times 10^{-9},$$

$$\varepsilon_\rho \leq 1.48948308787786630 \times 10^{-9},$$

$$\varepsilon_F \leq 1.20354325050525201 \times 10^{-9}.$$

Independent recomputation from the decimal CSV gives, respectively,

```
7.441921424927986e-09
1.4894832079072834e-09
1.203543250505252e-09
```

The tiny difference in the first two values is output-decimal roundoff. These are invariant and constraint errors, not residuals of all 18 ODEs.

For the ODE test, the checker differentiates the 171-point output with the five-point centered stencil and compares it with an independently implemented right-hand side. The largest component RMS residual is

$$4.705850657056059 \times 10^{-4},$$

below the declared **1e-3** acceptance bound. The largest normalized spatial Einstein finite-difference residual is

$$5.702272371471661 \times 10^{-7},$$

below **1e-6**. Finite-difference truncation at output spacing 0.01 dominates these residuals, so they must not be reported as **1e-9** equation accuracy.

The canonical replay is byte-identical in both `history.csv` and `summary.json`. The tighter run uses `rtol=1e-12`, `atol=1e-14`, and maximum step 0.001; it takes 2,294 solver steps and 2,422 right-hand-side evaluations. Its maximum normalized state difference from the canonical run is

$$6.310776406656671 \times 10^{-10}.$$

Independent recomputation gives refined maximum relative errors

$$(\varepsilon_S, \varepsilon_\rho, \varepsilon_F) = (1.6415774988196702 \times 10^{-9}, 3.2855796418252515 \times 10^{-10}, 3.31034411227924 \times 10^{-10}),$$

all smaller than their canonical counterparts.

5 Equation-by-equation numerical treatment

Each equation family is solved as part of the same implicit coupled system:

1. **a** equation: the nonlinear stage equation includes the product **a** **H**; it is advanced in the same Newton solve as every other component.
2. **H** equation: nonlinear scalar equation coupled through **rho(S)** and **p(S)**.
3. Eight spinor pairs (**psi_j**, **psi_(j+8)**): each has the exact two-component rotation/dilution form in Section 1, while all pairs remain nonlinearly coupled through the shared condensate **S**.

No equation is decoupled or solved by shortcut substitution; all 18 components are advanced simultaneously in each implicit Newton solve. The canonical initial state leaves six spinor pairs identically zero, but their equations are still present and checked. Only the (1,9) and (8,16) pairs become nonzero.

6 Dark-energy-like and dark-matter-like provenance

Within the implemented potential there are exactly two homogeneous background correspondences:

1. Dust-like component (candidate dark-matter analog):

$$\rho_{\text{dust}} = \frac{1}{20}S, \quad p_{\text{dust}} = 0.$$

1. Negative-pressure component (candidate dark-energy analog):

$$\rho_{\text{neg}} = \frac{19}{20}S^{1/5}, \quad p_{\text{neg}} = -\frac{4}{5}\rho_{\text{neg}}.$$

1. Total equation-of-state trajectory:

$$w = \frac{p}{\rho},$$

which evolves from less negative to more negative effective pressure dominance as **S** dilutes.

1. Fraction diagnostics used by the solver output:

$$f_{\text{dust}} = \frac{(1/20)S}{\rho}, \quad f_{\text{neg}} = \frac{(19/20)S^{1/5}}{\rho}.$$

Interpretation guardrails:

1. These are model-internal effective-fluid analogies, not direct observational identifications.
2. Split signature $(4,4)$ is mathematically consistent in this framework but is not a standard Lorentzian cosmology.
3. Dust-like background dilution does not establish dark-matter clustering.
4. Negative background pressure does not establish perturbative stability or an observational dark-energy fit.
5. No compactification, particle-physics identity, quantum theory, or precision-fit cosmology is supplied.

Other spinor potentials, nonminimal curvature terms, torsion couplings, and inhomogeneous modes are mathematically possible but were not implemented. Accordingly, this is exhaustive only for the selected two-term potential and homogeneous ansatz.

7 Reproduction and machine-readable evidence

The tracked `evidence.json` records exact component pairings, source hashes, solver configuration, four exact model checks, 22 numerical checks, and dark-sector scope. The tracked `convergence.json` records all 24 replay/refinement checks. The Phase 7 checker pins both files by SHA-256, and the gate regenerates each file twice and requires byte identity.

7.1 Windows PowerShell

```
python refinement\phase7-x0-x7\build_evidence.py
.\scripts\verify_phase7_x0_x7_reports.ps1
```

7.2 Git Bash or WSL

```
python.exe refinement/phase7-x0-x7/build_evidence.py
bash ./scripts/verify_phase7_x0_x7_reports.sh
```